How the Arabidopsis Information Resource (TAIR) and AgBioData Consortium Contribute to the FAIRification of Arabidopsis Data

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The Arabidopsis Information Resource (www.arabidopsis.org) is a knowledgebase for the model plant species *Arabidopsis thaliana* that strives to make Arabidopsis data Findable, Accessible, Interoperable and Reusable (FAIR). Over the past 25 years, TAIR has evolved into a self-sustaining resource with over 40,000 global registered users. TAIR curators extract and annotate experimental data from the literature, primarily focusing on gene function and genetic data such as mutant alleles and phenotypes. Recently, TAIR spearheaded a first-of-its-kind community initiative to re-annotate the Arabidopsis genome, the first new structural annotation in almost a decade. TAIR is a member of the AgBioData Consortium (www.agbiodata.org), a network of agricultural related resources with a mission to consolidate standards and best practices for acquiring, displaying and reusing genomic, genetic and breeding (GGB) data. Our participation in AgBioData has yielded tangible outcomes like the development of guidelines for the Arabidopsis research community on how to make their published data more FAIR, broadly facilitating data and metadata submission, and adoption of recommended nomenclature standards applied to the new genome. This alliance of agricultural database professionals supports our goal of making Arabidopsis research data more FAIR to advance our understanding of plant gene function and ultimately improve agriculture. Some of the future directions for AgBioData include improving biocuration practices using enabling technologies such as AI and continued work towards sustainable funding for biocuration.

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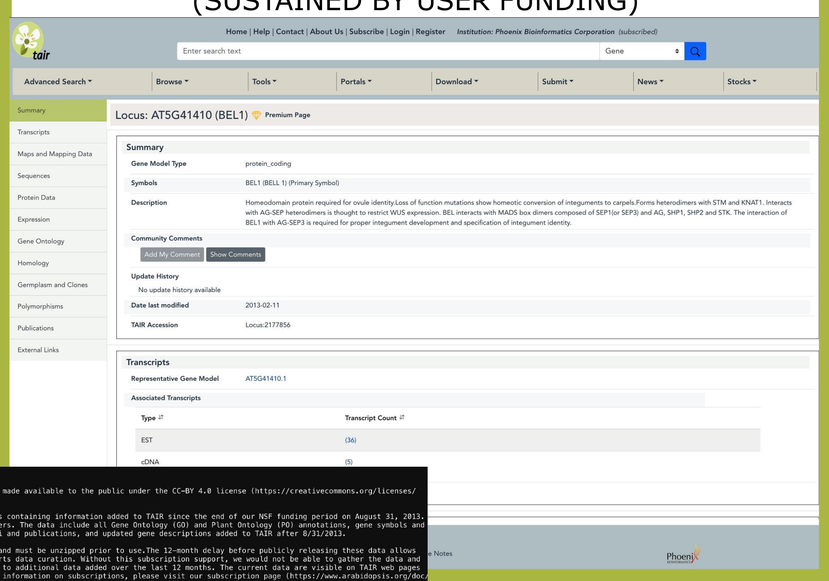
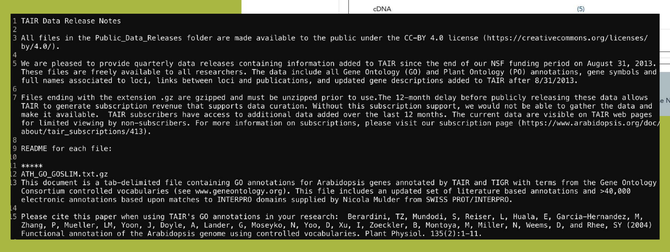
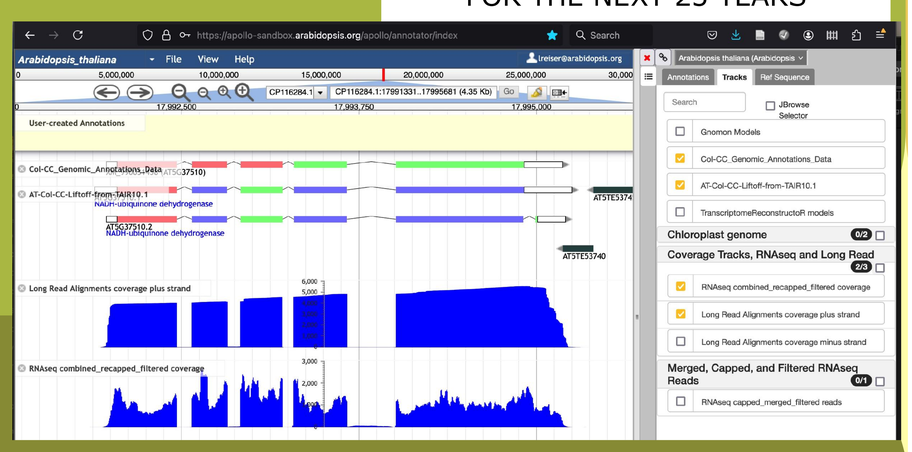
TAIR's KNOWLEDGEBASE AND TOOLS (SUSTAINED BY USER FUNDING)

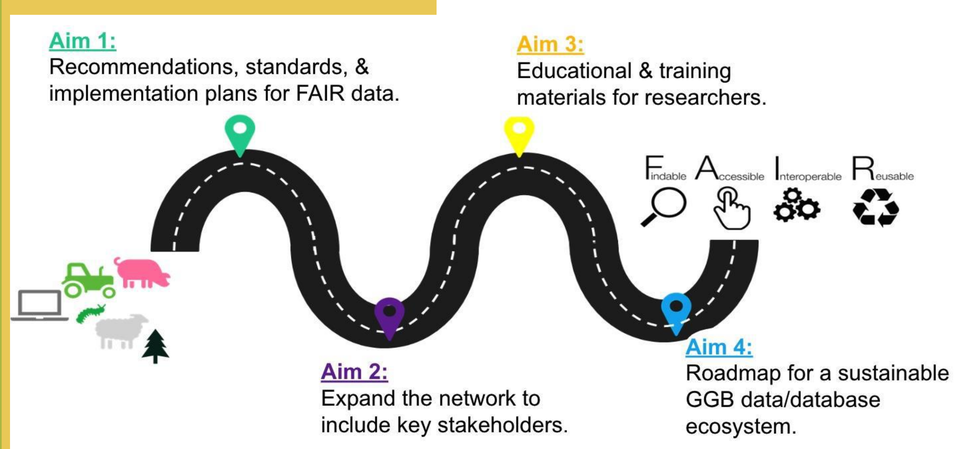




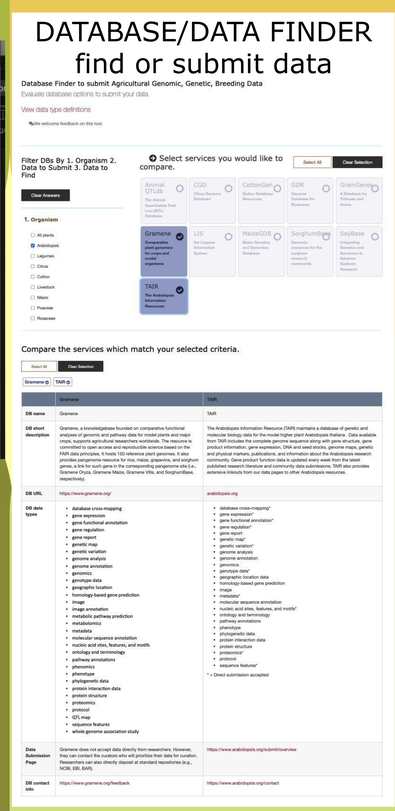
COMMUNITY CURATED T2T GENOME FOR THE NEXT 25 YEARS



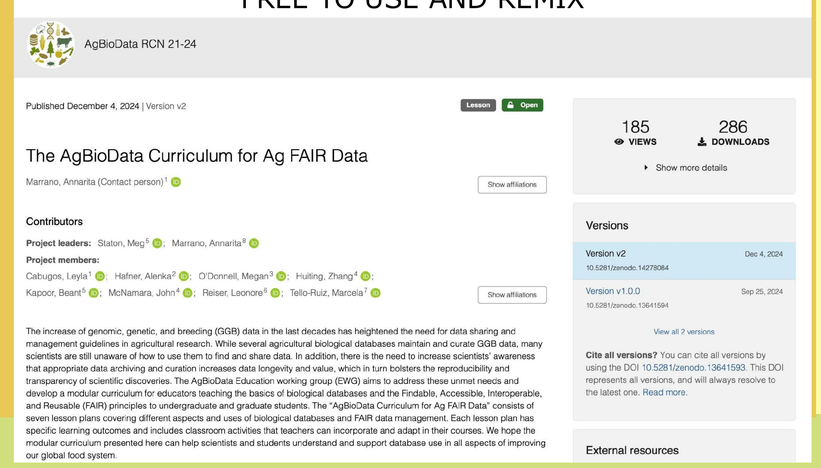
AGBIODATA RCN AIMS



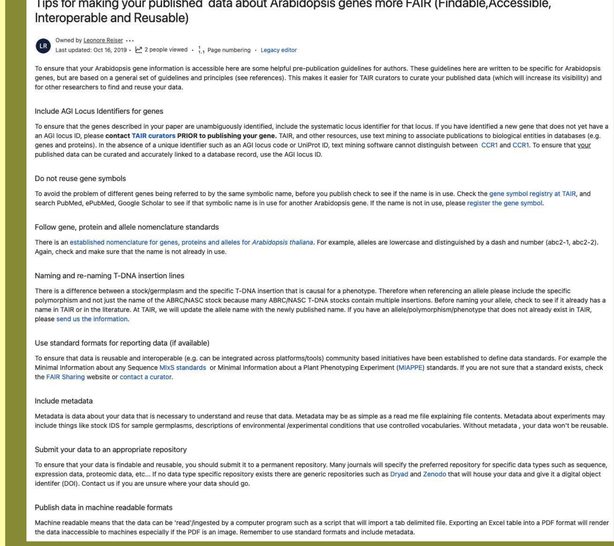
DATABASE/DATA FINDER find or submit data



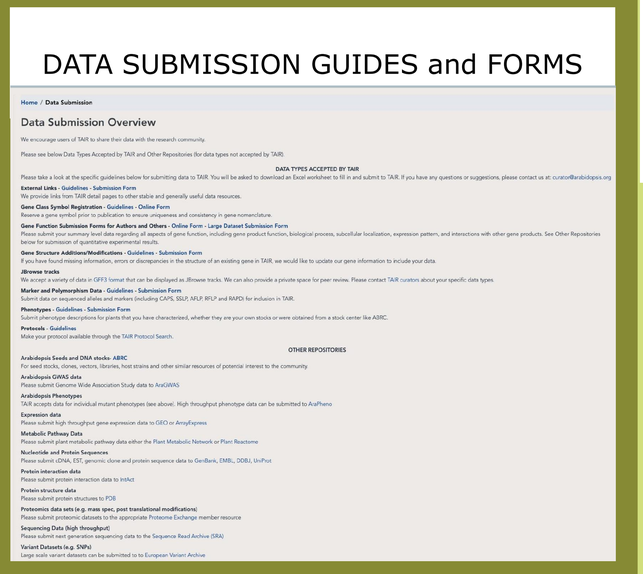
AGBIODATA FAIR AG DATA CURRICULUM FREE TO USE AND REMIX

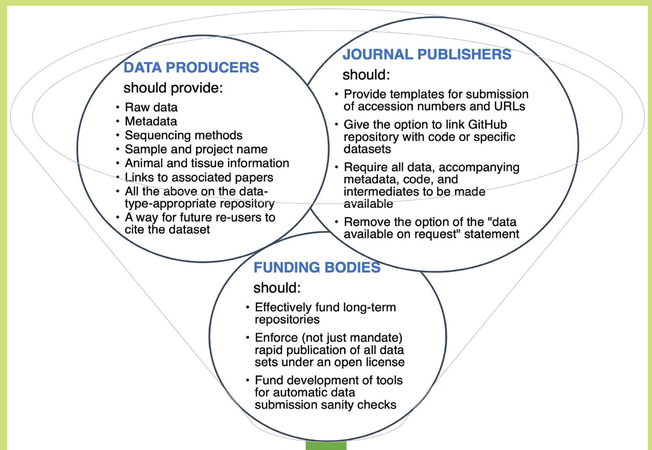


TAIR GUIDELINES FOR PUBLISHING FAIR ARABIDOPSIS DATA



DATA SUBMISSION GUIDES and FORMS





MORE DATA AVAILABLE FOR REUSE

WORKING GROUP GUIDELINES AND RECOMMENDATIONS

Recommendations for extending the GFF3 specification for improved interoperability of genomic data

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